

Keng Lin

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EDUCATION & Work Experience

PhD Candidate in Physics	Rutgers, The State University of New Jersey Jan. 2021 - Dec. 2026
Post-bachelor Student	Los Alamos National Laboratory Jan. 2020 - Dec. 2020
Research Assistant	Columbia University Nevis Laboratories Jan. 2019 - Jan. 2020
Bachelor of Art in Physics honor: <i>Cum laude</i>	Columbia University in the City of New York Graduated in Dec. 2018

Awards

URA visiting scholar, 07/01/2024 - 01/31/2025. Award:\$15,000

PUBLICATIONS & CONFERENCE TALKS & POSTERS

Selected Publications:

MiniBooNE Collaboration, "Updated MiniBooNE Neutrino Oscillation Results with Increased Data and New Background Studies", Phys. Rev. D 103,052002

MicroBooNE Collaboration, "The MicroBooNE Single-Photon Low-Energy Excess Search", MICROBOONE-NOTE-1087-PUB

CONFERENCE TALKS & POSTERS

Talks:

"Searches for Physics Beyond the Standard Model with the Short-Baseline Near Detector", [Lepton-Photon 2025](#)

"MicroBooNE's Beyond Standard Model Physics Program" , [NuFact 2024](#)

"Current and Future Perspectives of MicroBooNE's BSM Program" , [APS April Meeting 2024](#)

"Toward a heavy axion selection in MicroBooNE using the NuMI beam", [APS April Meeting 2023](#)

"Constraining 'Dirt' Backgrounds for the MicroBooNE Single Photon Low Energy Excess Search", [New Perspectives 2020](#)

"Study Neutrinos using MiniBooNE Detector", [Los Alamos National Laboratory Symposium 2020](#)

"Boosted Decision Tree Based Single Electron Selection in MicroBooNE", [APS April Meeting 2019](#)

Posters:

Jun. 2022, "Event Selection Tools Targeting Single-Photon Events in The Short-Baseline Near Detector", [Neutrino 2022](#)

Jun. 2020, "Constraining 'Dirt' Backgrounds for the MicroBooNE Single Photon Low Energy Excess Search", [Neutrino 2020](#)

RESEARCH EXPERIENCE

Collaborator @ SBND (2021 - Present) Advisor: Professor Andrew Mastbaum

- [Instrumentation] Install and commission the Cosmic Ray Tagger system (CRT)
- [Analysis] Lead the heavy QCD axions analysis and co-lead the Millicharged particle analysis.
- [Software] Contributed to the analysis packages [sbncode](#); see [poster](#).

Collaborator @ MicroBooNE (2018 - Present) Advisors: Professor Andrew Mastbaum

- [Analysis] Lead the heavy QCD axion analysis.
- [Simulation] Develop [TwoBodyDecayGen](#) module.
- [Software] Develop tools for MeV scale physics analysis.

Collaborator @ MicroBooNE Advisor: Professor Georgia Karagiorgi

- [Analysis] Study external event contribution to the single-photon LEE analysis; see [poster](#).
- [Analysis] Search for BNB single-electron using boosted decision tree (BDT)

Collaborator @ MiniBooNE (2020) Advisors: Dr. William Louis and Dr. Richard Van De Water

- [Analysis] Analyzed the extended MiniBooNE dataset (18.75×10^{20} POT); contributed to the 2020 publication, [Phys. Rev. D 103,052002](#)

Research Assistant @ Coherent CAPTAIN-Mills(CCM) experiment (2020)

Advisors: Dr. William Louis and Dr. Richard Van De Water

- [Hardware] Decommission CCM120 (first generation detector).

Research Assistant @ Columbia University (2018) Advisor: Professor Yuri Levin

- [Simulation] Implement an event simulator of the collision of a black hole and a hypothetical cosmic string.

TEACHING EXPERIENCE

- Teaching Assistant @ Rutgers

Advanced General Physics (Fall 2025) , General Physics (Fall 2025) , General Physics Lab (Summer 2025) , General Physics Lab (Spring 2021)

- Instructor of Material Science and Nanotechnology @ Columbia (Summer 2019)

PUBLIC OUTREACH

- Science Demonstrator at The Franklin Institute, Philadelphia, PA (2016)

SKILLS

Programing Languages

C, C++, Python, Mathematica, ROOT	- Proficient
JavaScript, Siemens NX	- Working Experience
R, MatLab, LabVIEW	- Exposure

Environment/Tools

Software:	LArSoft, G4BNB, Pythia8, PyHF
Applications:	Git, MicroSoft Office
Markup Languages:	HTML, XML, Markdown, $\LaTeX$
Operating Systems:	Linux, OS X, Windows

SIDE PROJECTS

3D Printing Detectors: MicroBooNE



and SBND



A map for neutrino experiments: [NeutrinoExperimentsOnEarth](#)